

AMS 563 PROJECT

Multimodal Transformer for Joint CT Imaging and EHR-based Risk Prediction

Abstract

This project explores the development of a multimodal deep learning framework for clinical risk prediction by jointly leveraging medical imaging and structured Electronic Health Record (EHR) data. Specifically, the work investigates how computed tomography (CT) imaging features can be combined with patient clinical variables to improve predictive modeling of medical outcomes. The proposed approach utilizes transformer-based architectures to enable feature fusion between radiological image embeddings and tabular EHR data, allowing the model to capture complementary patterns across modalities. The project also involves designing preprocessing pipelines for CT imaging data and structured clinical datasets, as well as exploring representation learning strategies for integrating heterogeneous healthcare data sources. By combining computer vision techniques with clinical data modeling, this work aims to evaluate whether multimodal learning can enhance predictive performance and support more informed clinical risk assessment.

Problem Statement

Clinical risk prediction models traditionally rely on structured Electronic Health Record (EHR) data such as laboratory values, demographics, and comorbidity indicators. While these variables capture important clinical signals, they often fail to incorporate rich information contained in medical imaging data.

Computed tomography (CT) scans contain detailed spatial and anatomical information that may provide additional insights into disease progression and patient risk profiles. However, integrating imaging data with structured clinical variables presents several challenges, including differences in data modalities, representation formats, and feature dimensionality.

This project investigates whether **multimodal deep learning models can improve clinical risk prediction by jointly leveraging CT imaging and structured EHR data**. Specifically, the research explores transformer-based architectures for feature fusion, enabling the model to learn relationships between radiological image embeddings and patient clinical variables. The goal is to evaluate whether combining imaging and tabular clinical data can enhance predictive performance compared to models trained on a single modality.